

PERSONAL

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EDUCATION

- **PhD in Biostatistics** | *Université Paris Cité, France* | 2019 – 2023
Pierre Louis Doctoral School of Public Health (ED 393 – Sorbonne Université)
 - **Thesis title:** “Development of a virtual biopsy system and diagnosis automation in kidney transplantation”
 - **Distinction:** Highest Honors (Très honorable), awarded unanimously by the jury
 - **Jury committee:** Umberto Maggiore, Rachel Patzer, Candice Roufousse, Xavier Jouven, Alexandre Loupy (supervisor)
- **MPH (Master 2) in Methodology and Statistics in Biomedical Research** | *Université Paris-Saclay, France* | 2018 - 2019
- **BSc in Mathematics** | *University of Alberta (Music Minor), Canada* | 2012 - 2014
- **BSc in Mathematics (Transfer)** | *State University of New York (SUNY) Plattsburgh, USA* | 2010 - 2012

CURRENT EMPLOYMENT

- Postdoctoral researcher under supervision of Prof. Olivier Jolliet at Technical University of Denmark (DTU) since 08/2024
 - <https://orbit.dtu.dk/en/persons/daniel-yoo>

FORMER EMPLOYMENTS

- Postdoctoral researcher under supervision of Prof. Emmanuel Mignot at Stanford University (co-supervision) (01/2024 – 07/2024)
 - <https://www.mignotlab.com/>
- Postdoctoral researcher under supervision of Prof. Alexandre Loupy at INSERM U970, Université Paris Cité (co-supervision) (01/2024 – 07/2024)
 - <https://paristransplantgroup.com/>
- Researcher PhD student under supervision of Prof. Alexandre Loupy, Director of team 11 at HEGP research center INSERM U970, Université Paris Cité (11/2019 – 10/2023)
- Data Scientist internship under supervision of Prof. Alexandre Loupy, Director of team 11 at HEGP research center INSERM U970, University Paris-Descartes (9/2018 – 10/2019)
- Management Information System Consultant retired as Sergeant at Republic of Korea Air Force (ROKAF) Headquarter, Gyeryong, Chungcheongnam-do, South Korea (7/2016 – 7/2018)

- Contract Research Assistant / R Lecturer at Korea Consumer Agency (KCA), Eumseong-gun, Chungcheongbuk-do, Korea (4/2016 – 6/2016) under supervision of Dr. Insuk Kim
- Data Analyst at Transcriptome Sciences Inc. (TSI), Edmonton, Alberta, Canada (6/2015 – 11/2015) under supervision of Prof. Philip F. Halloran and Dr. Jeff Reeve

PUBLICATIONS

1st author

MERWACS: Development and External Validation of a Non-Invasive Machine Learning Tool for Identifying Subject to be Screened for CKD

Yoo D., Nguyen V., Maggiore U., Jolliet O.
In Press in PLOS Digital Health

Enhancing Prediabetes and Diabetes Detection Through a Machine Learning-Enabled Self-Assessment Approach

Yoo D., Maggiore U., Jolliet O. Journal of Clinical Epidemiology, 2026; 195.

DOI: <https://doi.org/10.1016/j.jclinepi.2026.112266>

(2024 Impact Factor: 5.2, #16/102 in Health Care Sciences & Services)

Artificial intelligence for forecasting live donor kidney function

Yoo D., Maggiore U. J Nephrol 2024 July 29.

DOI: <https://doi.org/10.1007/s40620-024-02040-4>

(2023 Impact Factor: 2.7, #36/126 in Urology & Nephrology)

A machine learning-driven virtual biopsy system for kidney transplant patients

Yoo D.*, Divard G.*, Raynaud M., Cohen A., Mone T., Rosenthal J., Bentall A., Stegall M., Naesens M., Zhang H., Wang C., Gueguen J., Kamar N., Bouqueneau A., Batal I., Coley S., Gill J., Oppenheimer F., Souza E., Kuypers D., Durrbach A., Seron D., Rabant M., Huyen J., Campbell P., Shojai S., Mengel M., Bestard O., Jukic N., Jurić I., Boor P., Cornell L., Alexander M., Coates P., Legendre C., Reese P., Lefaucheur C., Aubert O., Loupy A. **Nature Communications** 2024 Jan 16; Nat Commun 15, 554 (2024).

DOI: <https://doi.org/10.1038/s41467-023-44595-z>

(2023 Impact Factor: 14.7, #8/134 in Multidisciplinary Sciences)

An automated histological classification system for precision diagnostics of kidney allograft

Yoo D.*, Goutaudier V.*, Divard G., Gueguen J., Astor B., Aubert O., Raynaud M., Demir Z., Hogan J., Weng P., Smith J., Garro R., Warady B., Zahr R., Sablik M., Twombly K., Couzi L., Berney T., Boyer O., Duong-Van-Huyen J.P., Giral M., Alsadi A., Gourraud P., Morelon E., Quintrec M., Brouard S., Legendre C., Anglicheau D., Villard J., Zhong W., Kamar N., Bestard O., Djamali A., Budde K., Haas M., Lefaucheur C., Rabant M., Loupy A. **Nature Medicine** 2023 May 04; Nat Med 29, 1211–1220 (2023).

DOI: <https://doi.org/10.1038/s41591-023-02323-6>

(2023 Impact Factor: 58.7, #1/313 in Biochemistry & Molecular Biology)

COVID-19 Pandemic Consequences On Worldwide Organ Transplantation : a population-based study

Aubert O.*, **Yoo D.***, Zielinski D.*, Cozzi E.*, Cardillo M., Dürr M., Gil B., Coll E., Silva M., Sallinen V., Lemström K., Midtvedt K., Ulloa C., Immer F., Weissenbacher A., Vallant N., Jukic N., Tanabe K., Papatheodoridis G., Menoudakou G., Torres M., Soratti C., Krogh D., Lefaucheur C., Ferreira G., Junior H., Hartell D., Forsythe J., Mumford L., Reese P., Kerbaul F., Jacquelinet C.*, Vogelaar S.*, Papalois V.*, Loupy A*. **Lancet Public Health** 2021 Aug 30; S2468-2667(21)00200-0.

DOI: [https://doi.org/10.1016/S2468-2667\(21\)00200-0](https://doi.org/10.1016/S2468-2667(21)00200-0)

(2021 Impact Factor: 72.4, #1/182 in Public, Environmental & Occupational Health)

*equal contribution

Co-author

Machine learning assessment of individual fatty acids and related dietary factors on all-cause mortality risk

Rebelo T., Huang L., Verly E., Zhao B., Nguyen V., Slavin J., Shamir R., **Yoo D.**, Jolliet O. **Food Research International**. Published May 1, 2026.

DOI: <https://doi.org/10.1016/j.foodres.2026.118628>

(2024 Impact Factor: 8.0, #16/172 in Food Science & Technology)

Microvascular Inflammation of Kidney Allografts and Clinical Outcomes

Sablik M., Sannier A., Raynaud M., Goutaudier V., Divard G., Astor B.C., Weng P., Smith J., Garro R., Warady B.A., Zahr R.S., Twombly K., Dharnidharka V.R., Dandamudi R.S., Fila M., Huang E., Sellier-Leclerc A.L., Tönshoff B., Rabant M., Verine J., del Bello A., Berney T., Boyer O., Catar R.A., Danger R., Giral M., **Yoo D.**, Girardin F.R., Alsadi A., Gourraud P.A., Morelon E., Le Quintrec M., Try M., Villard J., Zhong W., Bestard O., Budde K., Chauveau B., Couzi L., Brouard S., Hogan J., Legendre C., Anglicheau D., Aubert O., Kamar N., Lefaucheur C., Loupy A. **New England Journal of Medicine**. Published October 24, 2024.

DOI: <https://doi.org/10.1056/NEJMoa2408835>

(2023 Impact Factor: 96.3; #2/329 in Medicine, General & Internal)

Clinical Prediction Model for Antibody-Mediated Rejection: A Strategy to Minimize Surveillance Endomyocardial Biopsies After Heart Transplantation

Coutance G., Kransdorf E., Aubert O., Bonnet G., Yoo D., Rouvier P., Duong Van Huyen J.P., Bruneval P., Taupin J.L., Leprince P., Varnous S., Kobashigawa J., Jouven X., Patel J., Loupy A. **Circulation: Heart Failure**.

2022;0:CIRCHEARTFAILURE.122.009923

DOI: <https://doi.org/10.1161/CIRCHEARTFAILURE.122.009923>

(2022 Impact Factor: 9.7; #13/142 in Cardiac & cardiovascular systems)

Dynamic prediction of renal survival among deeply phenotyped kidney transplant recipients using artificial intelligence: an observational, international, multicohort study

Raynaud M., Aubert O., Divard G., Reese P.P., Kamar N., Yoo D., Chin C.S., Bailly É., Buchler M., Ladrière M., Quintrec M.L., Delahousse M., Juric I., Basic-Jukic N., Crespo M., Silva H.T., Linhares K., Castro M.C.R., Pujol G.S., Empaña J.P., Ulloa C., Akalin E., Böhmig G., Huang E., Stegall M., Bentall A., Montgomery R., Jordan S., Oberbauer R., Segev D., Friedwald J., Jouven X., Legendre C., Lefaucheur C., Loupy A. **Lancet Digital Health** 2021 October 27; S2589-7500(21)00209-0.

DOI: [https://doi.org/10.1016/S2589-7500\(21\)00209-0](https://doi.org/10.1016/S2589-7500(21)00209-0)

(2021 Impact Factor: 36.615; #1/31 in medical informatics; #9/172 in medicine, general & internal)

Trajectories of glomerular filtration rate and progression to end stage kidney disease after kidney transplantation.

Raynaud M., Aubert O., Reese P.P., Bouatou Y., Naesens M., Kamar N., Bailly É., Giral M., Ladrière M., Le Quintrec M., Delahousse M., Juric I., Basic-Jukic N., Gupta G., Akalin E., Yoo D., Chin C.S., Proust-Lima C., Böhmig G., Oberbauer R., Stegall M.D., Bentall A.J., Jordan S.C., Huang E., Glotz D., Legendre C., Montgomery R.A., Segev D.L., Empana J.P., Grams M.E., Coresh J., Jouven X., Lefaucheur C., Loupy A. *Kidney Int.* 2021 Jan;99(1):186–197. **Kidney Int.** 2021 Jan;99(1):186-197.

DOI: <https://doi.org/10.1016/j.kint.2020.07.025>

(2021 Impact Factor: 18.998; #3/90 in Urology & Nephrology)

Identification and characterization of trajectories of cardiac allograft vasculopathy after heart transplantation: a population-based study

Loupy A., Coutance G., Bonnet G., Van Keer J., Raynaud M., Aubert O., Bories M.C., Racapé M., Yoo D., Duong Van Huyen J.P., Bruneval P. *Circulation.* 2020 Jun 16;141(24):1954-67.

DOI: <https://doi.org/10.1161/CIRCULATIONAHA.119.044924>

(2020 Impact Factor: 29.690; #3/142 in Cardiac & cardiovascular systems; #1/65 in Peripheral Vascular Disease)

Archetype analysis identifies distinct profiles in renal transplant recipients with transplant glomerulopathy associated with allograft survival

Aubert O., Higgins S., Bouatou Y., Yoo D., Raynaud M., Viglietti D., Rabant M., Hidalgo L., Glotz D., Legendre C., Delahousse M. *Journal of the American Society of Nephrology.* 2019 Apr 1;30(4):625-39.

DOI: <https://doi.org/10.1681/ASN.2018070777>

(2019 Impact Factor: 9.274; #5/85 in Urology & Nephrology)

Deceased donor kidney transplantation across donor-specific antibody barriers: predictors of antibody-mediated rejection

Schwaiger E., Eskandary F., Kozakowski N., Bond G., Kikić Ž., Yoo D., Rasoul-Rockenschaub S., Oberbauer R., Böhmig G.A. *Nephrology Dialysis Transplantation.* 2016 Aug 1;31(8):1342-51;

DOI: <https://doi.org/10.1093/ndt/gfw027>

(2016 Impact Factor: 4.470; #4/25 in Transplantation)

PEER REVIEW ACTIVITIES

- PLOS Digital Health
 - Reviewed 2 manuscripts in 2026
- Nature Sustainability
 - Reviewed 1 manuscript in 2025
- Journal of Nephrology
 - Reviewed 1 manuscript in 2026
 - Reviewed 1 manuscript in 2025
 - Reviewed 2 manuscripts in 2024
- Transplantation reviews
 - Reviewed 1 manuscript in 2026

- Reviewed 2 manuscripts in 2024

CONFERENCES PRESENTATION (ORAL COMMUNICATION)

- Helsinki (Finland) November 11-14, 2025, European Public Health Conference (EPH): *Estimating years of life lost and gained from dietary risks using an accelerated failure time model*
- Helsinki (Finland) November 11-14, 2025, European Public Health Conference (EPH): *Enhancing prediabetes and diabetes detection through a machine learning-enabled self-assessment*
- Helsinki (Finland) November 11-14, 2025, European Public Health Conference (EPH): *Development and Validation of a Machine Learning Model for Reduced Kidney Health Risk Assessment*
- Philadelphia (USA) June 1-5, 2024, American Transplant Congress (ATC): *Development and Validation of Antibody-Mediated Rejection Prediction Model on Augmented Data Using Generative Model in Kidney Transplant Patients*
- Athens (Greece) September 17-20, 2023, The European Society for Organ Transplantation (ESOT): *Development and validation of antibody-mediated rejection prediction model on augmented data using generative model in kidney transplant patients*
- Athens (Greece) September 17-20, 2023, The European Society for Organ Transplantation (ESOT): *Development, application, and validation of a histological classification automation system for kidney allograft precision diagnostics*
- Milan (Italy) June 15-18, 2023, European Renal Association (ERA): *Development, application, and validation of a histological classification automation system for kidney allograft precision diagnostics*
- San Diego (USA) June 3-7, 2023, American Transplant Congress (ATC): *Development, application, and validation of a histological classification automation system for kidney allograft precision diagnostics*
- Lyon (France) December 13-16, 2022, Société Francophone de Transplantation (SFT): *Développement, application et validation d'un système automatisé de la classification histologique de Banff en transplantation rénale*
- Rennes (France) October 4-7, 2022, Société Francophone de Néphrologie Dialyse et Transplantation (SFNDT): *Développement, application et validation d'un système automatisé de la classification histologique de Banff en transplantation rénale*
- Boston (USA) June 4-8, 2022, American Transplant Congress (ATC): *Development and validation of a machine learning based virtual biopsy system in kidney transplant patients*
- Boston (USA) June 4-8, 2022, American Transplant Congress (ATC): *Development of a comprehensive Banff Automation System for kidney allograft precision diagnostics*
- Paris (France) May 19-22, 2022, European Renal Association (ERA): *Development and validation of a machine learning based virtual biopsy system in kidney transplant patients*
- Geneva (Switzerland) 09/12/2021, Société Francophone de Transplantation (SFT): *Développement and validation d'un système de biopsie virtuelle en transplantation rénale*
- Toulouse (France) 06/10/2021, Société Francophone de Néphrologie Dialyse et Transplantation (SFNDT): *Développement and validation d'un système de biopsie virtuelle en transplantation rénale*

- Milan (Italy) 01/09/2021, The European Society for Organ Transplantation (ESOT): *Development of a comprehensive Banff automation system for kidney allograft precision diagnostics*
- Milan (Italy) 01/09/2021, The European Society for Organ Transplantation (ESOT): *Development and validation of a virtual biopsy system in kidney transplant patients*
- California (USA) 12/12/2019, Artificial Intelligence in Medicine (AIMed): *Development and validation of a virtual biopsy system in kidney transplant patients*

CONFERENCES PRESENTATION (POSTER)

- Copenhagen (Denmark) November 6, 2025, Health Data Science Day: *Machine Learning Self-Assessment for Detecting Prediabetes and Diabetes*
- San Diego (USA) November 9-12, 2025, AIMed25: *An artificial intelligence-derived self-assessment early warning system of prediabetes or diabetes*
- San Diego (USA) November 9-12, 2025, AIMed25: *Development and Validation of a Machine Learning Model for Reduced Kidney Health Risk Assessment*
- Gothenburg (Sweden) May 13-15, 2025, 14th Nordic Public Health Conference (NPHC): *An AI-Derived Home-Based Early Warning System for Prediabetes or Diabetes*
- Banff (Canada) September 19-23, 2022, Banff-Canadian Society of Transplantation (CST): *Development, application, and validation of a histological classification automation system for kidney allograft precision diagnostics*

CONFERENCES CHAIR

- Helsinki (Finland) November 11-14, 2025, European Public Health Conference (EPH): *Digital Health and Artificial Intelligence*
- Athens (Greece) September 17-20, 2023, The European Society for Organ Transplantation (ESOT): *AI & Digital Health: from donation to the outcome*

SUPERVISION

PhD supervision

- Teresa Rebelo (Technical University of Denmark)
 - Machine Learning Assessment of Healthy Lifestyles with co-supervisions of Assistant Prof. Daniel Fozer and Prof. Olivier Jolliet

TEACHING

SPS (Specialpædagogisk Støtte, Special Educational Support)

- 5 students at Technical University of Denmark (DTU) with special need in 2026 – spring semester
- 3 students at Technical University of Denmark (DTU) with special need in 2025 – autumn semester
- 1 student at Technical University of Denmark (DTU) with special need in 2025 – spring semester

Lecture

- Guest Lecturer at University of Parma, Italy, May 7th, 2026 – 2 hours
 - Lectured medical doctors: Machine Learning in Medicine (Machine Learning fundamentals and clinical applications, ML model development pipeline, model validation and generalizability, clinical AI

- evaluation metrics, explainability methods, data quality and class imbalance, reproducibility in clinical ML research)
- R Lecturer at Korea Consumer Agency (KCA), Korea (4/2016 – 6/2016) – 10 hours
 - Lectured basic and intermediate levels of usage of statistical software R (Data Management, Data Wrangling, Data Visualization, Basic Programming, Statistical Analysis, Basic Machine Learning)

Mentoring

- Mentor at Danish Data Science Academy (DDSA), 2025 – 2026

CERTIFICATIONS

- [Professional Certificate in PyTorch for Deep Learning](#) (DeepLearning.AI)
- [Professional Certificate in Food, Nutrition, and Health](#) (Wageningen University & Research)
- [Healthy and Sustainable Foods and Products Specialization](#) (University of Michigan)
- [Human Health and Climate Change](#) (United Nations)
- [AI for Good Specialization](#) (DeepLearning.AI)
- [Professional Certificate in Food Security and Sustainability](#) (Wageningen University & Research)
- [Professional Certificate in Animal Breeding and Genetics](#) (Wageningen University & Research)
- [Professional Certificate in Fundamentals of Google AI for Web Based Machine Learning](#) (Google)
- [Big Data for Agri-Food: Principles and Tools](#) (Wageningen University & Research)
- [Machine Learning Engineering for Production \(MLOps\) Specialization](#) (DeepLearning.AI)
- [Professional Certificate in Tiny Machine Learning \(TinyML\)](#) (HarvardX)
- [Professional Certificate in Data Analysis for Genomics](#) (HarvardX)
- [Generative Adversarial Networks \(GANs\) Specialization](#) (DeepLearning.AI)
- [Professional Certificate in TensorFlow Developer](#) (DeepLearning.AI)
- [Deep Learning Specialization](#) (DeepLearning.AI)
- [Writing in the Sciences](#) (Stanford University)
- [Climate Change: Carbon Capture and Storage](#) (The University of Edinburgh)
- [Machine Learning](#) (Stanford University)
- [Professional Certificate in Data Science](#) (HarvardX)
- [Mathematics for Machine Learning Specialization](#) (Imperial College London)
- [Data Science Specialization](#) (Johns Hopkins University)
- [Oracle Certified Professional, Java SE 6 Programmer](#) (Oracle)
- [Data Mining Specialization](#) (University of Illinois at Urbana-Champaign)
- [Big Data XSeries](#) (University of California, Berkeley)

POPULAR MEDIA

- [Grefe de rein : un algorithme pour éviter les faux diagnostics de rejets](#) (Kidney transplant: an algorithm to avoid false diagnosis of rejection) (*Le Figaro*)
- [Un algorithme améliore le diagnostic de rejet de greffe du rein](#) (Algorithm improves diagnosis of kidney transplant rejection) (*Le Monde*)
- [Automating kidney transplant diagnostics](#) (*Nature Medicine*)

- [Huge decrease in organ transplants as Covid took hold across world](#) (*The Guardian*)
- [Organ transplants down by a third](#) (*The Times*)
- [Organ transplants 'fell by a third during pandemic' as Covid delayed operations](#) (*The Daily Telegraph*)
- [La pandémie a entraîné un net ralentissement des transplantations d'organes](#) (The pandemic has led to a marked slowdown in organ transplants) (*Le Monde*)
- [Covid-19 : les greffes d'organes en baisse de près de 16% entre 2019 et 2020](#) (Covid-19: organ transplants down nearly 16% between 2019 and 2020) (*Le Figaro*)
- [Organ Transplants Fell By A Third Worldwide During First Covid-19 Wave](#) (*Forbes*)
- [Organ transplants saw marked decline worldwide amid pandemic](#) (*New York Post*)
- [Organ transplants saw marked decline worldwide amid pandemic: study](#) (*Fox News*)

MEMBERSHIPS

- 2026 – Current: Ingeniørforeningen i Danmark (Danish Society of Engineers; IDA)
- 2025 – Current: Danish Data Science Academy (DDSA)
- 2024 – Current: AI task force at DTU Sustain
- 2023 – 2024: The European Society for Organ Transplantation (ESOT)
- 2022 – 2024: Building Resilience Against crisis: a systematic and global approach to adVancE organ Safety and supply in Transplantation (BRAVEST) project funded by European Union (EU4Health)
 - <https://www.bravest-project.eu/>
- 2022 – 2024: European Renal Association (ERA)
- 2021 – 2024: Société Francophone de Transplantation (SFT)

PRIZES / GRANTS / BURSARY

- 2023: The European Society for Organ Transplantation (ESOT), Digital Transformation in Transplantation Best Abstract Award
- 2023: American Transplant Congress (ATC), Young Investigator Award
- 2022-2024: Building Resilience Against crisis: a systematic and global approach to adVancE organ Safety and supply in Transplantation (BRAVEST) project funded by European Union (EU4Health), handles WP3 as Biostatistician and Data Scientist
 - <https://www.bravest-project.eu/>
- 2022: European Renal Association (ERA), the top 10 best abstracts (2nd place)
- 2021: The European Society for Organ Transplantation (ESOT), Full Bursary to best abstracts

LANGUAGES

- Korean and English fluently
- French, Japanese, Danish basic